map-charting-student-math-misunderstandings

# ✨ MAP 6th Place Solution - Qwen-semble FTW! 🚀

| | |
|---|---|
| **Rank** | 6 |
| **Team** | Manan Jhaveri |
| **Author** | Manan Jhaveri |
| **Topic ID** | 612099 |
| **Version** | Original |

Source: https://www.kaggle.com/competitions/map-charting-student-math-misunderstandings/discussion/612099
Retrieved with Kaggle CLI on 2026-10-21.

Thanks to the organizers and Kaggle for this fun and challenging competition. Also thanks to the participants for all the valuable notebooks and discussion posts.

# Highlights of my solution (TL;DR version)

- I used a mix of `Qwen3-Embedding-8B`, `Qwen3 14B` and `Qwen2.5 14B`.
- Synthetic data for rare labels, Augmented text and Pseudo-labelling duplicates helped to improve the score.

# Detailed solution

## Problem statement

- We are provided with math questions, answer given by students and the explanations given by the students.
- We need to predict 3 things:
  - Is the answer correct or not
  - Is the explanation correct, has a misconception or neither
  - What is the type of misconception, if present

## Data and pre-processing

- From the start, I saw this problem as just a misconception classification problem. So I removed the `True_` and `False_` prefixes from the categories and used the resulting 37 labels during fine-tuning.
- Used the following input format:

```
correctness =   row[]
input_text = (




)
```

In the initial experiments, I noted that adding the last line of the prompt was improving CV and LB scores for the LLMs. Hence, decided to keep it for the rest of the experiments.

- I dropped the duplicate samples altogether. Duplicates = same answer and explanation but different misconception as label.
- In the last few experiments, I pseudo-labelled these duplicates and added them back to the training data.

- In some experiments, I added ~50 augmented samples per category. Augmentation involved basic replacement of punctuations, conversion of words to number or vice versa, replacing "plus" to "+", replacing "equals to" to "equals" or "=", adding small typos, etc.
  - I had analyzed near duplicates found through fuzzy search in the training data and accordingly tried to create augmented samples.
- In the final few experiments, I added synthetic data for 10/37 categories with the lowest support. I used ChatGPT 4o to generate 100-120 samples per category.
- I used a max sequence length of 160 with dynamic padding. This made the training and inference faster for me.
- For local testing, I have used 20% test split and used full data for re-training the model for the submissions.

## Models

- I started my experiments with a variety of models like GTE, Ettin and later moved to LLMs of up to 8B parameters. I observed that the `sequence classification` approach with `Qwen3-Embedding-8B` performed the best for me.
- After freezing my prompt template and pre-processing, I moved to 14B param models too.
- I trained these LLMs with 4-bit Q-LoRA. Rank = 128, alpha = 32-128 (as per different experiments).
- I used A5000/A100 on Jarvis Labs for compute.

Here's the summary of the models that I finally used in my qwen-semble:

| Model | Data used | Public LB |
|---|---|---|
| Qwen3 Embedding 8B | Deduplicated data | 0.946 |
| Qwen3 14B | Deduplicated data | 0.947 |
| Qwen2.5 14B IT | Deduplicated data | 0.947 |
| Qwen3 Embedding 8B | Deduplicated data + Augmented data | 0.945 |
| Qwen3 Embedding 8B | Deduplicated data + Pseudo-labelled duplicates + Synthetic data | **0.949** (best single model) |
| Qwen3 14B | Deduplicated data + Pseudo-labelled duplicates + Synthetic data | 0.948 |
| Qwen3 14B | Deduplicated data + Pseudo-labelled duplicates + Synthetic data (+ different seed than above) | 0.943 |

## Ensembling

- I used the ensembling method from this notebook by @kishanvavdara : https://www.kaggle.com/code/kishanvavdara/ensemble-gemma-qwen-deepseek
- The ensemble of the first 4 models from the above table helped me crack 0.95 on the public LB.
  - I used this ensemble to pseudo-label the duplicates.

    - The model with augmented data has a poor individual score compared to the rest but it contributed significantly to the ensemble. I tried removing it multiple times as I got better models but the score always dipped.
- With the ensemble of the first 6 models, I got 0.951 on the public LB (and 0.947 private LB).
- In the last 2 days, I was out of ideas and I didn't want to make small tweaks to model weightage in the ensemble method as it felt unreliable. So I decided to fine-tune Qwen3 14B with the same config as the one I used earlier in the 0.948 model, but with a different seed and random state. Its individual score was quite poor but ensembling it with the other 6 models gave me my best public as well as private LB score.

## Failed experiments

- Multiple pre-trained models.
- Several prompt templates.
- Adding too many augmented samples. I had shortlisted multiple patterns from my analysis and created ~10k augmented samples. But the public LB score remained ~0.945 and these models didn't add much to the ensemble.
- Different modelling approaches:
    - Using all original 65 labels.
    - I tried a 2-stage approach where the first stage model just predicts whether a student explanation is correct, has a misconception or neither of the two. And the second stage would predict the misconception type.
    - Text Generation approach where I provided simplified descriptions of each category (that a question can possibly have) in the prompt and asked the model to predict which one best describes the reasoning behind the student's explanation.
    - Textual Entailment approach where I tried to predict whether a given sample entails any of the simplified descriptions of the misconception categories.
    - MoE-style NN for the classification head instead of LoRA adapters. (really wanted this to work but couldn't give enough time to fix the hyperparams)
- Different loss functions like focal-loss